



THE UNIVERSITY
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University of
Southampton



Research Software Maintenance Fund

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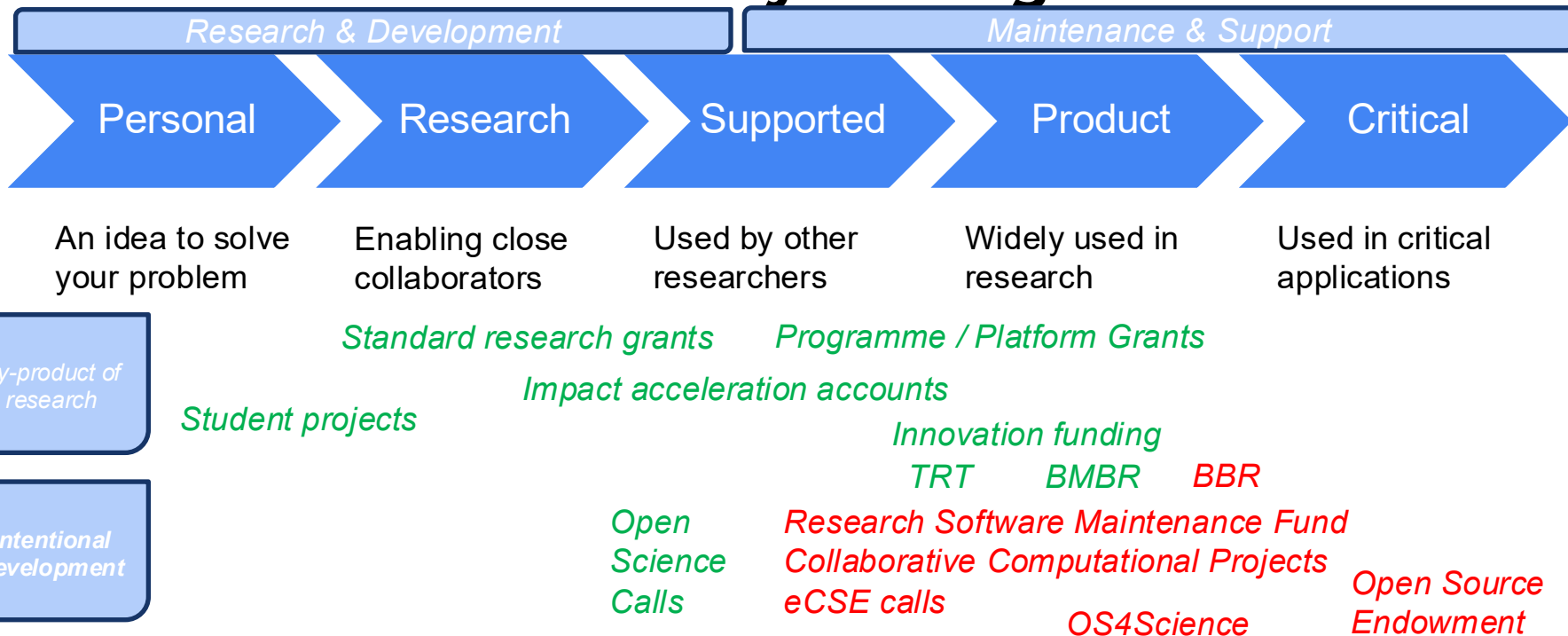


UK Research
and Innovation

History of research software development funding

- Originally (pre-1990s), research software was a by-product of investment into computational research and hardware ([10.17226/6323](#))
 - Much software developed in research grants, maintained tacitly
 - Exception was the Collaborative Computational Projects ([10.1002/pro.3298](#))
- The 2000's saw large-scale government investment in e-Science driven by increase in data ([10.1007/s00287-019-01215-9](#))
 - Focus on middleware, but across all disciplines ([10.12759/hsr.37.2012.3.147-164](#))
 - Opportunities for productisation / development through incubators
- Then came the RSE movement (2012-present), and more funding for maintenance, software-specific calls to build capability
 - RSE Fellowships, Software for the Future, Tools and Resources Development Fund
 - Increase in philanthropic funding to support specific problem areas (e.g. life sciences)
- Now comes GenAI and digital sovereignty

Research Software Lifecycle Stages



A Framework for Understanding Research Software Sustainability (2021) ([10.5281/zenodo.4988277](https://doi.org/10.5281/zenodo.4988277))

Overview of research software funding landscape (2022) ([10.5281/zenodo.6102487](https://doi.org/10.5281/zenodo.6102487))

Research Software Maintenance Fund

£4.8m fund run by the Software Sustainability Institute on behalf of UKRI

Aims:

- Understand how to better fund software maintenance, including what types of activity should be prioritised for funding;
- Support key software used by the community; and
- Diversify the people and software supported to encourage a wider range of contributions. (*could include international, industry, charities, etc.*)

Two rounds of funding, the first with large awards (up to £500k for two years) and small awards (up to £150k for one year), second only small awards.

Two stage process: expressions of interest and full applications

Supports UK research (in any discipline), multiple stable releases, used outside of dev team

Extremely high levels of interest

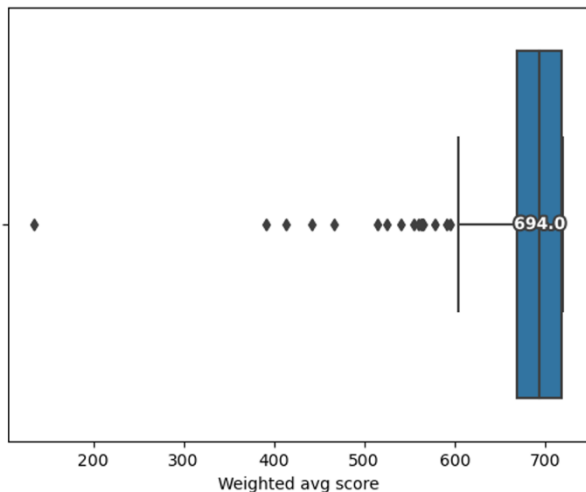
- Round 1: 360 expressions of interest, 150 full applications invited, 143 submitted, 4 large / 9 small projects funded
- Round 2: 229 expressions of interest, 61 full applications invited, 60 submitted, expect to fund ~10 small projects

Szilard Point – the point at which the total value of the time spent by applicants / reviewers / administrators exceeds the total value of the funding call ([10.1038/d41586-025-04060-x](https://doi.org/10.1038/d41586-025-04060-x))

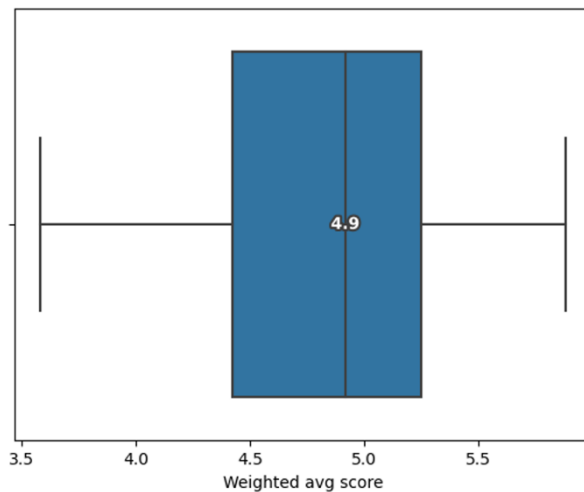
- Revised EoI assessment process for Round 2, using Distributed Peer Review and splitting domain areas into separate pools.

Extremely high fit to criteria

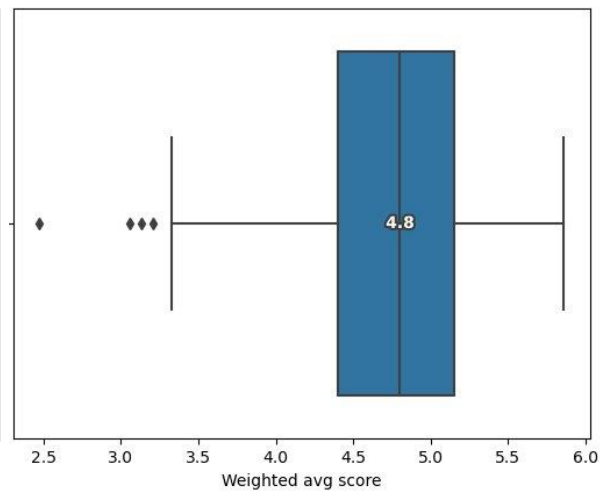
Criteria: eligibility, vision and impact, feasibility and approach, capability to deliver



Round 1: only 7 of 360 EoIs out of scope, only 1 completely out of scope



Round 1: only 11 of 143 full applications below threshold of 4 (should be funded), none below 3 (could be funded)



Round 2: only 18 of 229 EoIs below threshold after DPR and moderation

Szilard Point Analysis

Round 1

- 360 expressions of interest (10 hours / Eol) = 3,600 hours
- 720 Eol reviews (0.5 hours / review) = 360 hours
- 150 full applications (40 hours / application) = 6,000 hours
- 450 full application reviews (1 hours / review) = 450 hours
- Review panel (60 reviewers) = 420 hours
- *Call management (6 months @ 0.5 FTE) = 412 hours*

Value of effort: ~11,000 hours = ~6.7 person years = ~£1m
Value of call: ~£3m

Reducing the number of full applications is the primary way of reducing cost of effort. Doesn't factor in reuse of work.

Round 2

- 229 expressions of interest (10 hours / Eol) = 2,290 hours
- 1,100 Eol reviews (0.5 hours / review) = 550 hours
- 60 full applications (40 hours / application) = 2,400 hours
- 240 full application reviews (1 hour / review) = 900 hours
- Review panel (47 reviewers) = 329 hours
- *Call management (6 months @ 0.5 FTE) = 412 hours*

Value of effort: ~6,500 hours = ~3.9 person years = ~£588k
Value of call: ~£1.8m

Research Areas

	R2 Small Eols	R2 Small Eols %	R1 Small Eols	R1 Small Eols %	R1 Large Eols	R1 Large Eols %
Physical, Environmental, Mathematical & Computer Science and Engineering	139	60.7%	236	62.9%	223	60.6%
<i>Physical & Environmental Sciences</i>			97	25.9%	77	20.9%
<i>Engineering & Technology</i>			51	13.6%	55	14.9%
<i>Mathematical & Computer Sciences</i>			88	23.5%	91	24.7%
Life Sciences	46	20.1%	74	19.7%	76	20.7%
Social Sciences	22	9.6%	31	8.3%	31	8.4%
Arts and Humanities	18	7.9%	15	4.0%	15	4.1%
Other	4	1.7%	19	5.1%	23	6.3%

Applicants could choose multiple research areas in R1 but were asked to choose a main area in R2

Types of activity requested (R1 Eol)

Almost all the Eol's included some form of technical activities.

Large awards were more likely to include Community and Governance activities than small awards

	Total	Large	Small
Technical	99%	99%	99%
Community	81%	87%	75%
Documentation	84%	88%	81%
Training	65%	69%	62%
Governance	27%	35%	20%

Job role of lead

	R1 Awarded %	R1 Invited %	R1 Rejected %	R1 EoI Submitted %	R2 EoI Submitted %
Academic / Researcher	77%	61%	59%	60%	67%
Research Technical Professional (including Research Software Engineer)	23%	25%	24%	24%	24%
Research Facilitator / Research Manager		4%	5%	4%	4%
Project Manager / Product Manager		11%	12%	12%	8%

Applicants could choose multiple roles at EoI stage. Of these, the most common pairing was “Academic / Researcher and Research Technical Professional (including Research Software Engineer).”

Key takeaways for funders

- The volume of applications evidences the need for more programmes to support sustainability of software
- More focussed schemes may be more effective and efficient however general schemes may have higher engagement
- Additional support to improve the ability of some groups (e.g. RSEs/RTPs, AHSS) to submit high-quality application is needed - level of institutional support / knowledge also varies
- It is extremely difficult to provide meaningful individual feedback when large numbers of applications are scoring highly – and you will get hate mail either way
- Lots of discussion in the metascience community around more efficient ways of funding, e.g. DPR, lottery approaches, nominations

Key takeaways for applicants

- Understanding what you need funding for – then you can reuse previous work on applications
- Being able to demonstrate the impact of your software on research is the key differentiator
- Know the landscape that your software operates in – what other software, data and services are used with it, what are your competitors?
- The highest scoring RSMF applications offered multiple ways to engage with developers and help provide input into the roadmap.
- Research software funders are increasingly coordinating
- **Funding is, unfortunately, very much driven by luck**